

Muhammad Asif Ali Khan

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MS Data Science graduate (Silver Medalist, CGPA 3.64/4.0) specializing in federated learning, with two papers under review at the Springer journal Machine Learning on fair, contribution-aware aggregation. My research focuses on quantifying client contribution in federated systems using Shapley-value-based credit assignment, and designing aggregation strategies that remain robust and fair under non-IID data and heterogeneous client quality. Seeking to pursue a PhD extending this work toward robust, privacy-preserving, and theoretically-grounded federated learning.

RESEARCH AREA

Federated Learning

Title: A Contribution-Aware Aggregation in Federated Learning

- Architected a contribution-aware Federated Learning framework to improve model stability under non-IID data and partial client participation.
- Integrated FedProx-style proximal local training with FW-Shapley contribution estimation and Beta-Shapley coalition sampling for scalable, efficient client utility measurement.
- Designed dynamic aggregation weights to amplify high-utility client updates and suppress low-utility/harmful ones.
- Evaluated on MNIST and Fashion-MNIST, achieving smoother convergence and improved generalization over FedAvg, FedProx, SFedAvg, and GreedyFed.

Journal Article (under review)

Title: SimFL-FWShapley: Efficient and Drift-Aware Shapley-Based Credit Assignment for Federated GBDT Learning

Journal: Springer journal Machine Learning

Title: FedCAWS: An Efficient Amortized Learning-Based Contribution-Aware Aggregation Framework for Federated Learning

Journal: Springer journal Machine Learning

EDUCATION

MS Data Science | Shifa Tameer-e-Millat University, Islamabad

May 2024 – March 2026

- Silver Medalist
- **Relevant Coursework:** Machine Learning, Statistical and Mathematical Methods for Data Science, NLP, Data Distribution Processes, Tools and Techniques, Computer Vision, Deep Learning
- **CGPA:** 3.64/4.0

BS Mathematics | International Islamic University Islamabad

Sep 2019 – Oct 2023

- **Relevant coursework:** Linear Algebra, Calculus, Topology, Ring Theory, Software Tool, Theory of Module, Differential Geometry, Functional Analysis, Complex Analysis
- **CGPA:** 3.05/4.0

SKILLS

Programming Languages: Python, R

ML/DL Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn, NLTK

NLP: Sentiment Analysis, Text Classification, LangChain, Word Embeddings (Word2Vec, GloVe, TF-IDF), Transformers (BERT, GPT)

Deep Learning: CNN, RNN, LSTM, Transformers

Computer vision: YOLOv5

Data Analytics & Visualization: Pandas, NumPy, PyCaret, Seaborn, Matplotlib, Plotly, ggplot

Tools: Jupyter Notebook, Google Colab, VS Code

EXPERIENCE

AI Engineer Intern | Eziline Software House | Islamabad Pakistan

Feb 2026 – July 2026

- Built and deployed a Flask-based Machine Learning web app for real-time classification of abusive vs. non-abusive Roman Urdu comments using NLP, text preprocessing, and TF-IDF features, achieving 88% classification accuracy across multiple trained classifiers.
- Designed a 2D U-Net deep learning model for brain tumor segmentation on multi-modal MRI scans, achieving a Dice score of 0.85, with preprocessing, IoU evaluation, prediction overlay visualization, and a Streamlit UI.
- Developed a Streamlit-based NLP application for real-time emotion detection from English text, achieving 90% model accuracy with confidence-score prediction.
- Engineered and deployed an AI Voice Assistant web app using Streamlit, GroqCloud LLM, Whisper-based Speech-to-

Text, and gTTS Text-to-Speech, enabling browser-based voice interaction with real-time audio responses.

- Built a real-time object detection pipeline using PyTorch and YOLOv5s (fine-tuned on PASCAL VOC, 20 classes), with automated preprocessing, annotation conversion, webcam/video inference, and ONNX/TorchScript export for edge deployment readiness.

PROJECTS

Book Recommendation System | Shifa Tameer-e-Millat University

Aug 2024

- Created a hybrid book recommendation system combining collaborative and content-based filtering, using NLP for text preprocessing and feature extraction from book descriptions; deployed via Flask/Streamlit for personalized, interactive recommendations.

Bank Churn Prediction using Deep Learning

Oct 2025

- Implemented a deep neural network to predict customer churn on real-world bank data with 88% accuracy, using SHAP for explainable AI (XAI) and feature importance analysis to improve model interpretability.

House Price Prediction

June 2025

- Trained a regression model to predict house prices on the Kaggle dataset, applying feature engineering and hyperparameter tuning to achieve a high R^2 score using Gradient Boosting and Random Forest.

Student Stress Level Classifier

Jan 2026 – March 2026

- Constructed a multi-class ML model (Logistic Regression) to classify student stress levels (Low/Medium/High) with 90% accuracy, deployed via a Streamlit dashboard with real-time predictions, confidence scores, and personalized health recommendations.
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